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标题	In Situ-Generated Niobium-Catalyzed Synthesis of 3-Pyrroline Derivatives via Ring-Closing Metathesis Reactions
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摘要（原文节选）

An active in situ-generated Nb complex was used as a catalyst in the ring-closing metathesis reaction of N,N-diallyl-p-toluenesulfonamide to afford the corresponding 3-pyrroline derivative. The Nb complex was formed from NbCl5, trimethylsilyl chloride, Zn, and PhCHCl2 in tetrahydrofuran. The Nb complex displayed high catalytic activity toward ring-closing metathesis reactions. ⋯

数据表 · Table 1（24 行，含数值行 19 行）

Optimization of Reaction Conditionsa

entry	catalyst precursor	additive	conv. (%)b	yield (%)b
1	NbCl5	PhCHCl2	>99	>99 (94)
2c	NbCl5	PhCHCl2	>99	>99
3	NbCl4(THF)2	PhCHCl2	>99	>99
4	NbCl3(DME)	PhCHCl2	>99	>99
5d	NbCl3(DME)	PhCHCl2	>99	>99
6	Nb(OEt)5	PhCHCl2	91	91
7	NbF5	PhCHCl2	14	5
8	TaCl5	PhCHCl2	9	nd
9	TiCl4	PhCHCl2	nr	nd
10	FeCl3	PhCHCl2	nr	nd
11	none	PhCHCl2	nr	nd
12e	NbCl5	PhCHCl2	nr	nd
13f	NbCl5	PhCHCl2	75	60
14g	NbCl5	PhCHCl2	1	trace
15h	NbCl5	PhCHCl2	nr	nd
16i	NbCl5	PhCHCl2	nr	nd
17j	NbCl5	PhCHCl2	nr	nd
18j	NbCl4(THF)2	PhCHCl2	nr	nd
19k	NbCl5	PhCHCl2	53	50
20l	NbCl5	PhCHCl2	56	4
21m	NbCl5	PhCHCl2	>99	>99
22n	NbCl5	PhCHCl2	>99	>99
23	NbCl5	PhCH2Cl	nr	nd
24	NbCl5	PhCCl3	nr	nd

脚注（逐字）：aReaction conditions: 1a (1 mmol), catalyst (0.1 mmol), trimethylsilyl chloride (TMSCl, 0.5 mmol), Zn (1.5 mmol), additive (0.5 mmol), and THF (3 mL) at 60 ° C for 2 h.bConversions and yields were determined by gas chromatography (GC); the number in parentheses shows the isolated yield.cNbCl5 (5 mol

正文对应句（逐字）：The RCM of N,N-diallyl-p-toluenesulfonamide was initially investigated as a model substrate to optimize the reaction conditions (Table 1). When a mixture of niobium pentachloride,8 trimethylsilyl chloride (TMSCl), Zn, benzyl dichloride, and N,N-diallyl-p-toluenesulfonamide in tetrahydrofuran (THF) w

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